

VEDAANG CHOPRA

vedaangchopra@gatech.edu — +1 (404) 740-9905 — Atlanta, GA — [LinkedIn](#) — [GitHub](#) — [Website](#)

MS CS (ML) at Georgia Tech — Advised by **Dr. Matthew Gombolay**, CORE Robotics Lab

Research: Production agentic AI systems – Multi-agent orchestration, planning & tool use – LLM evaluation – RAG

EDUCATION

Georgia Institute of Technology

Atlanta, GA

M.S. in Computer Science (*Machine Learning Specialization*), **GPA: 4.0/4.0**

Aug 2024 – Dec 2026

– **Coursework:** Agentic AI, Deep Learning, Deep RL, ML Security, Large & Vision Language Models, Systems for AI

Maharaja Surajmal Institute of Technology

New Delhi, India

B.Tech. in Information Technology, CGPA: 8.8/10.0

Aug 2016 – Aug 2020

AI RESEARCH & ENGINEERING EXPERIENCE

Georgia Institute of Technology – CORE Robotics Lab @ Siemens

Atlanta, GA

Graduate Research Assistant — Advisor: Dr. Matthew Gombolay

Jan 2026 – Present

- Building a closed-loop agentic pipeline for parametric CAD code generation: an LLM plans and generates CadQuery programs, an execution environment returns compile and geometry feedback, and a LangGraph-orchestrated repair loop iteratively corrects errors – an autonomous agent workflow with verifiable execution reward across a modular multi-package system.
- Own the full evaluation lifecycle: pytest harness with golden datasets, regression suites, and VLM-as-judge visual scoring gating every change; ablations isolating retrieval, prompt, and repair-policy contributions to quality.

Georgia Institute of Technology – CS 8903 Special Problems

Atlanta, GA

Graduate Researcher

Jan 2025 – Present

- **SHASTRA** (Jan 2026–Present): Building an agent orchestration framework with constraint-aware plan search over cost, latency, and accuracy budgets; component registry with pluggable storage; and a runnable trace-to-graph pipeline extracting reusable workflow templates from prior agent execution traces.
- **ATHENA** (Jan–May 2025): Built a multi-agent system with supervisor-worker orchestration via LangGraph Command routing; coordinated 5 specialized agents (ideation, character design, world building, story generation, scene breakdown) plus an intent-planning agent, with a reflection router that critiques outputs, re-routes with targeted feedback, and dynamically modifies plans.
- **ARTEMIS** (Aug 2025–Present): Developing a cost-aware routing system for Vision-Language Models that dynamically selects the optimal VLM per request; trained a neural multi-task router with SLA-aware load balancing (simulation-validated, 90.3% oracle-utility recovery), routing across 5 VLMs (10 vLLM endpoints) spanning VQA, OCR, captioning, and reasoning tasks.

INDUSTRY EXPERIENCE

Fortinet Technologies Inc. (AIOps R&D)

Bengaluru, India

Software Development Engineer I & II (ML/AI)

Feb 2021 – Jul 2025

- Led development of an agentic RAG diagnostics system: LLM plans multi-step tool calls over network telemetry, invokes APIs, and verifies hypotheses via structured function calling behind prompt-level output guardrails, producing autonomous root-cause analysis – reduced mean resolution time 70% (hackathon prototype to production deployment).
- Scaled OpenSearch ingestion pipelines from 50 to 2,000 events/sec using Golang while maintaining reliability, observability, and production monitoring; integrated ML models into backend services with Python and Go across distributed enterprise infrastructure.
- Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection using density-based analysis; reduced manual troubleshooting 75%.

PUBLICATIONS & PATENTS

Patent (Filed): PCT/IN2022/058026 – Unsupervised distributional thresholding for wireless connectivity anomaly detection
Ontology-based Text Classification (Structured Knowledge Representation) — **Published: CEUR Workshop Pro**

TECHNICAL SKILLS

Agentic AI: LangChain, LangGraph, Multi-Agent Orchestration, Supervisor-Worker Patterns, Planning & Reasoning, Tool/Function Calling, Agentic RAG, Agent Memory, Reflection & Self-Critique, LLM-as-a-Judge, Structured Generation (Pydantic)

RAG / Knowledge: Semantic Search, FAISS, Vector Databases, Cross-Modal Retrieval, SBERT, CLIP

Evaluation: Offline Benchmarks, A/B Experimentation, Regression Suites, Human Evaluation, Confidence Estimation

AI / ML: PyTorch, Hugging Face Transformers, Deep Learning, NLP, Computer Vision, Multimodal AI, Scikit-Learn, Small Language Model Fine-Tuning (Edge Assistant)

LLMs / RL: vLLM, VLM Routing, Inference & Model Optimization (ONNX), PPO/DQN (academic projects)

Systems: Python, Go, C/C++, SQL/PostgreSQL, Redis, OpenSearch/Elasticsearch, Docker, CI/CD, Linux, Git, ONNX Runtime, FastAPI, Azure (no AWS/GCP), W&B